

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM  
(AUTONOMOUS)**

**II B.Tech I Semester Regular Examinations, November-2024**

**ADVANCED DATA STRUCTURES AND ALGORITHM ANALYSIS  
(Common to CSE, IT, CSE(AI&ML), CSE(DS), CSE(AI) and CSE(CS))**

**Time: 3 hours**

**Max. Marks: 70**

**Note:**

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

| Q. No.         | Questions                                                                                                                                                     | BTL | CO  | Marks |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|
| 1.             | a) Differentiate between best, average, and worst case complexities.                                                                                          | K1  | CO1 | 2M    |
|                | b) What is a Height balancing in AVL tree?                                                                                                                    | K1  | CO1 | 2M    |
|                | c) Describe the three main steps in Quicksort.                                                                                                                | K1  | CO2 | 2M    |
|                | d) Write the Applications of priority queues.                                                                                                                 | K1  | CO2 | 2M    |
|                | e) Write Control Abstraction of Greedy method.                                                                                                                | K1  | CO3 | 2M    |
|                | f) What are the searching methods that are commonly used in branch and bound method?                                                                          | K1  | CO3 | 2M    |
|                | g) Write about the constraints and criterion function used in backtracking.                                                                                   | K1  | CO4 | 2M    |
|                | h) Why Backtracking always produces an optimal solution?                                                                                                      | K1  | CO4 | 2M    |
|                | i) What is meant by NP-complete problem?                                                                                                                      | K1  | CO5 | 2M    |
|                | j) Define Chromatic Number in view of NP Hard graph problem.                                                                                                  | K1  | CO5 | 2M    |
| <b>UNIT-I</b>  |                                                                                                                                                               |     |     |       |
| 2.             | i) Insert the following sequence of elements into an AVL tree, starting with an empty tree: 10, 20, 15, 25, 30, 16, 18, 19.<br>ii) Delete 30 in the AVL tree. | K3  | CO1 | 10M   |
| <b>OR</b>      |                                                                                                                                                               |     |     |       |
| 3.             | a) Create a 2-3 tree from the following list of data items 5,6,8,21,12,30,34,27,23,4,33,7,24,9,10,11,13,38.                                                   | K3  | CO1 | 5M    |
|                | b) Explain with example, what are the different cases followed while inserting a node in 2-3-Tree.                                                            | K2  | CO1 | 5M    |
| <b>UNIT-II</b> |                                                                                                                                                               |     |     |       |
| 4.             | a) Derive the time complexity of Merge sort in average case.                                                                                                  | K3  | CO2 | 5M    |
|                | b) Explain about strassen's multiplication with an example.                                                                                                   | K2  | CO2 | 5M    |
| <b>OR</b>      |                                                                                                                                                               |     |     |       |
| 5.             | a) Define the following terms i)connected components ii)Bi connected components                                                                               | K2  | CO2 | 5M    |
|                | b) Write the applications of DFS & BFS.                                                                                                                       | K2  | CO2 | 5M    |



| UNIT-III |    |                                                                                                                                                                                                                                 |    |     |     |
|----------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----|
| 6.       | a) | Explain Kruskal's Minimum cost spanning tree algorithm with suitable example.                                                                                                                                                   | K2 | CO3 | 5M  |
|          | b) | Discuss the time and space complexity of Dynamic Programming traveling sales person algorithm.                                                                                                                                  | K2 | CO3 | 5M  |
| OR       |    |                                                                                                                                                                                                                                 |    |     |     |
| 7.       |    | Describe the Dynamic 0/1 Knapsack Problem. Find an optimal solution for the dynamic programming 0/1 knapsack instance for $n=3$ , $m=6$ , profits are $(p_1, p_2, p_3) = (1, 2, 5)$ , weights are $(w_1, w_2, w_3) = (2, 3, 4)$ | K3 | CO3 | 10M |
| UNIT-IV  |    |                                                                                                                                                                                                                                 |    |     |     |
| 8.       | a) | Draw the portion of state space tree for 4 queen's problem using variable - tuple sized approach.                                                                                                                               | K3 | CO4 | 5M  |
|          | b) | Relate Hamiltonian cycle with travelling sales person problem and also give the backtracking solution vector that finds all Hamiltonian cycles for any directed or undirected graph.                                            | K3 | CO4 | 5M  |
| OR       |    |                                                                                                                                                                                                                                 |    |     |     |
| 9.       | a) | Explain the importance of bounding function in generating the solutions. And also write about different types of bounding functions with an example each.                                                                       | K2 | CO4 | 5M  |
|          | b) | Explain the Graph-coloring problem. And draw the state space tree for $m=3$ colors $n=4$ vertices graph. Discuss the time and space complexity.                                                                                 | K2 | CO4 | 5M  |
| UNIT-V   |    |                                                                                                                                                                                                                                 |    |     |     |
| 10.      | a) | Explain the P, NP, NP-Hard and NP- complete classes with suitable examples.                                                                                                                                                     | K2 | CO5 | 5M  |
|          | b) | State and prove Cook's theorem.                                                                                                                                                                                                 | K2 | CO5 | 5M  |
| OR       |    |                                                                                                                                                                                                                                 |    |     |     |
| 11.      | a) | What are the differences between NP-Hard and NP-complete classes? Explain with examples.                                                                                                                                        | K2 | CO5 | 5M  |
|          | b) | Write about the theory of NP-completeness                                                                                                                                                                                       | K2 | CO5 | 5M  |